

Sai Udiga - CTF Platform Setup and Operations Guide

Prepared for: ****Sai Udiga****

System: ****Lloyds CTF / CTF-Main****

Repository: <https://github.com/saiudiga110/CTF-Main>

Local path: /Users/rislab/CTF-Main

1. Quick Answer

For normal event hosting, you should not need to configure the cluster manually again. From Mac mini 1, run:

```
cd /Users/rislab/CTF-Main
./scripts/ctfctl.sh start
```

Then open:

? Player site: <http://10.34.204.246>
? Admin: <http://10.34.204.246/admin>
? Architecture view: <http://10.34.204.246/plugins/ctfd-target/admin/architecture>
? Operations: <http://10.34.204.246/plugins/ctfd-target/admin/ops>

Use this to verify health:

```
./scripts/ctfctl.sh doctor
```

2. Current Machine Roles

Machine	Hostname	LAN IP	Role
Mac mini 1	rislab-mini1.local	10.34.204.246	Control plane + local worker
Mac mini 2	rislab-mini2.local	192.168.10.52	Remote Docker worker
Mac mini 3	rislab-mini3.local	192.168.10.53	Remote Docker worker

Mac mini 1 runs CTFd, MariaDB, Redis, proxy, Instance Manager, one local worker agent, and SSH tunnels to the remote worker agents. Mac mini 2 and Mac mini 3 run worker agents only. The tunnels are important on Docker Desktop because containers on Mac mini 1 may not route directly to the 192.168.10.x lab subnet.

3. Architecture

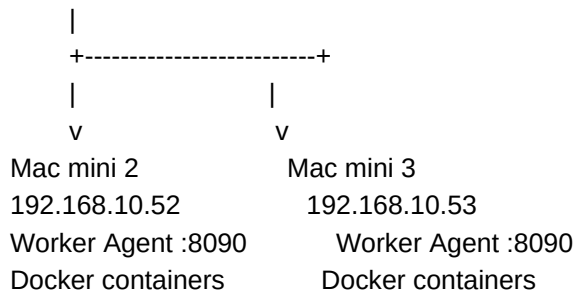
Players/Admin Browser

|
v

Mac mini 1 - 10.34.204.246

- Nginx proxy
- CTFd web application
- MariaDB
- Redis

- Instance Manager
 - Local Worker Agent
 - SSH tunnels:
- 127.0.0.1:18091 -> mac-mini-2:8090
 127.0.0.1:18092 -> mac-mini-3:8090



4. How Images Are Shared

Images are not pulled at player launch time. The startup command builds or verifies the required images on Mac mini 1 and then syncs current image IDs to Mac mini 2 and Mac mini 3.

Required workload images include:

- ? kali-ctf:latest
- ? vbank-ctf:latest
- ? vbank-auth:1
- ? vbank-idor:1
- ? vbank-logic:1
- ? vbank-xss:1
- ? vbank-staff-auth:1
- ? vbank-staff-deep:1
- ? vbank-analytics:latest

The startup script compares Docker image IDs. If a remote machine has an old or missing image, it sends a compressed Docker archive and loads it there. The same command also opens or reuses the SSH tunnels needed for the manager container to contact remote worker agents.

5. How Target Placement Works

When a player starts a target or Kali session:

1. The browser calls CTFd.
2. CTFd asks the Instance Manager on Mac mini 1.
3. The Instance Manager reads live worker heartbeats and capacity.
4. It chooses Mac mini 1, 2, or 3 based on available CPU/RAM/container/port capacity.
5. It tells that machine's worker agent to start the Docker container locally. For remote workers, the manager reaches the agent through the automatic SSH tunnel advertised as host.docker.internal:18091 or host.docker.internal:18092.
6. The selected worker returns a player URL like <http://192.168.10.52:20003/>.

If Mac mini 1 is full, new launches move to Mac mini 2 or Mac mini 3 automatically.

Target and Kali sidecars for the same player/challenge are colocated on the same worker so Kali can reach the target by vbank-app inside the shared Docker network.

6. Automation Commands

Run all commands from Mac mini 1:

```
cd /Users/rislab/CTF-Main
```

Start or resume everything

```
./scripts/ctfctl.sh start
```

Show status

```
./scripts/ctfctl.sh status
```

Show useful URLs

```
./scripts/ctfctl.sh urls
```

Run a full quick doctor check

```
./scripts/ctfctl.sh doctor
```

View local logs

```
./scripts/ctfctl.sh logs
```

Clean stale load-test containers and test records

```
./scripts/ctfctl.sh cleanup-tests
```

7. Installation From GitHub on a Fresh Mac mini 1

```
cd /Users/rislab
```

```
git clone https://github.com/saiudiga110/CTF-Main.git
```

```
cd CTF-Main
```

```
chmod +x start.sh scripts/start-macmini-cluster.sh scripts/ctfctl.sh
```

```
./scripts/ctfctl.sh start
```

Prerequisites:

? Docker Desktop installed and running on all three Mac minis.

? Passwordless SSH from Mac mini 1 to Mac mini 2 and Mac mini 3.

? Same LAN connectivity between the three machines.

? SSH local forwarding allowed from Mac mini 1 to each remote Mac mini; ctfctl start creates the required forwards automatically.

? Mac mini 2 hostname: rislab-mini2.local.

? Mac mini 3 hostname: rislab-mini3.local.

8. Available Endpoints

User and player endpoints

Endpoint Purpose

/ CTFd player site

/challenges Challenge list

/plugins/ctfd-target/target/start Start player target

/plugins/ctfd-target/target/status Target status

/plugins/ctfd-target/target/stop Stop target

/plugins/ctfd-target/kali/start Start Kali/noVNC sidecar

/plugins/ctfd-target/kali/status Kali status

/plugins/ctfd-target/kali/sso-url Kali noVNC access URL

/plugins/ctfd-target/kali/stop Stop Kali sidecar

/plugins/ctfd-target/health Public plugin health

Admin endpoints

Endpoint Purpose

/admin CTFd admin panel

/plugins/ctfd-target/admin/architecture Live architecture view

/plugins/ctfd-target/admin/architecture.json Architecture JSON for UI refresh

/plugins/ctfd-target/admin/ops Event operations dashboard

/plugins/ctfd-target/admin/network Network and DNS settings

/plugins/ctfd-target/admin/settings Target/Kali settings

/plugins/ctfd-target/admin/dashboard Running instance dashboard

Instance Manager endpoints

Endpoint Purpose

http://10.34.204.246:8088/health Manager health

/workers/heartbeat Worker registration heartbeat

/nodes Current fresh workers

/architecture Full architecture and inventory data

/instances Instance allocation/listing

/instances/<id>/status Instance status

/instances/<id> DELETE Stop instance

9. Production Notes

? The cluster is currently clean and healthy with 3 fresh nodes and zero test containers.

? Current image sizes are reduced: vBank about 170 MB and Kali about 2.85 GB.

? Architecture page auto-refreshes every 10 seconds.

? Worker heartbeats now retry after transient manager timeouts instead of exiting.
? Instance allocation is locked in the manager to avoid duplicate port assignment under concurrent launch bursts.
? DNS port 53 on Mac mini 1 is currently occupied by another local service. The app still works through direct IP: <http://10.34.204.246>.

10. Capacity Test Result

The current cluster reliably reached 21 Kali + 21 vBank target pairs in sequential testing before Docker create calls became too slow and timed out. The requested 30+ Kali sessions are close, but not yet reliable with on-demand Kali creation.

To make 30+ Kali sessions production reliable, use one or both of these improvements:

1. Pre-warm a pool of Kali containers before players arrive.
2. Reduce Kali startup cost further or split Kali sessions across more worker machines.

The automation is ready, but the platform should avoid sudden burst creation of 30+ Kali desktops at once unless a warm pool is implemented.

11. Troubleshooting

Check everything

```
./scripts/ctfctl.sh doctor
```

Architecture page does not show all 3 nodes

Run the startup command again. It refreshes worker containers, heartbeats, image sync, and SSH tunnels.

```
./scripts/ctfctl.sh start
```

Then wait 30 seconds and run:

```
./scripts/ctfctl.sh status
```

Remote worker is stale

Check SSH:

```
ssh rislab@rislab-mini2.local hostname  
ssh rislab@rislab-mini3.local hostname
```

Check tunnel listeners on Mac mini 1:

```
ls -l /dev/tcp/10.34.204.246/18091  
ls -l /dev/tcp/10.34.204.246/18092
```

Check Docker on remote:

```
ssh rislab@rislab-mini2.local 'docker ps'
ssh rislab@rislab-mini3.local 'docker ps'
```

Then restart the whole cluster:

```
./scripts/ctfctl.sh start
```

Stale test containers remain

```
./scripts/ctfctl.sh cleanup-tests
./scripts/ctfctl.sh status
```

DNS does not work

Use direct IP:

```
http://10.34.204.246
```

DNS port 53 is occupied on Mac mini 1. This does not block direct-IP operation.

Images are old on remote machines

Run:

```
./scripts/ctfctl.sh start
```

The startup script compares Docker image IDs and syncs changed images automatically.

12. Files Added or Changed for Operations

File Purpose

scripts/ctfctl.sh Operator command wrapper

scripts/start-macmini-cluster.sh Full cluster start/sync command

target-plugin/templates/architecture.html Architecture page UI

infra/instance_manager/app.py Manager scheduling, architecture API, allocation lock

infra/instance_manager/worker_agent.py Worker inventory and resilient heartbeat

infra/instance_manager/docker_worker.py Serialized Docker create operation

challenges/parrot/Dockerfile Reduced Kali image size

challenges/vbank-ctf/Dockerfile Reduced vBank image size

13. Daily Event Checklist

Before event:

```
cd /Users/rislab/CTF-Main
```

```
./scripts/ctfctl.sh start  
./scripts/ctfctl.sh doctor
```

During event:

? Keep Architecture open: <http://10.34.204.246/plugins/ctfd-target/admin/architecture>
? Keep Operations open: <http://10.34.204.246/plugins/ctfd-target/admin/ops>

After event:

```
./scripts/ctfctl.sh cleanup-tests
```

Stop player instances from the Operations page if needed.